

COURSE PROJECT

Project Proposal

1 Title & Team

Project Title	<i>Beyond University Rankings: A Transparent, Student-Centered Ranking of Global Sustainability Master's Programs</i>
Team Name	<i>Sustainability Matter</i>
Submission Date	<i>30 July 2026</i>

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2 Executive Summary

This project will build a transparent, searchable ranking of sustainability-focused Master's programs worldwide for prospective students, advisors, and SUSANhub users. Existing rankings often assess sustainability as part of broader institution-level performance, while students need program-level information on curriculum relevance, affordability, experiential learning, research strength, career signals, and peer perception. This project makes sustainability the primary lens by combining official program data, institutional sustainability indicators, research-output measures, and Reddit-based student discussion signals into a transparent ranking framework. By July 30, the team will deliver a cleaned program database, a documented scoring methodology, and a dashboard prototype that allows users to search programs, compare score components, and test alternative weighting scenarios.

3 Problem Statement

Prospective students interested in sustainability-focused Master's programs face a fragmented and inconsistent information environment. Program details are spread across university websites, brochures, tuition pages, ranking platforms, sustainability reports, employment pages, and online student communities. This makes it difficult for students to compare programs across countries, especially when they care about multiple factors such as curriculum focus, tuition, duration, experiential learning, research strength, career outcomes, and sustainability credibility.

The affected group is mainly prospective Master's students, especially international students comparing programs across regions. It also affects academic advisors, sustainability education networks, and SUSANhub users who need a free and transparent resource for sustainability education. The cost of leaving this problem unsolved is that students may over-rely on university reputation or broad institutional rankings, even though those rankings may not reflect the quality or fit of a specific Master's program. A university may perform strongly in sustainability overall, but a specific program may still be expensive, too theoretical, weak in experiential learning, or poorly aligned with a student's career goals.

Evidence of the problem can be seen in existing ranking structures. QS Sustainability and THE Impact Rankings evaluate universities, not individual Master's programs. AASHE STARS provides institution-level sustainability performance, while UN PRME identifies schools committed to responsible management education but does not compare specific programs. These sources are useful, but they do not directly answer the student's program-choice question.

This matters now because sustainability education is expanding, and students are increasingly seeking programs related to ESG, sustainable finance, climate strategy, responsible management, and social impact. Existing rankings often include sustainability as one component of broader institutional performance, while this project makes sustainability the primary lens at the program level. At the same time, peer discussion on Reddit and similar platforms shows that students want practical information about whether programs are “worth it,” how strong career outcomes are, and what current or former students think. A transparent, sustainability-first, program-level ranking can combine official data with peer sentiment to provide a more practical decision tool.

4 Objectives & Scope

Objectives	<i>Build a structured database of at least 50 sustainability-related Master’s programs in business or management, including program name, university, country, tuition, duration, curriculum focus, delivery mode, experiential learning, and sustainability indicators.</i> <i>Collect and analyze public Reddit posts and comments for at least 30% of the sampled programs or universities, using sentiment analysis and topic tagging to summarize peer perceptions.</i> <i>Develop a transparent scoring and ranking framework that combines program-level data, institutional sustainability indicators, research signals, affordability, career signals, and optional Reddit peer sentiment.</i> <i>Deliver a searchable dashboard prototype and final report, allowing users to filter programs, view ranking factors, and understand how final scores are calculated.</i>
In Scope	<i>Sustainability-related Master’s programs in business, management, ESG, sustainable finance, responsible management, social impact, climate strategy, and related fields.</i> <i>A selected global sample rather than a complete worldwide census.</i> <i>Public data collection from official university websites, rankings, institutional sustainability sources, OpenAlex, and Reddit.</i> <i>Data cleaning, normalization, missing-data flags, ranking-score calculation, sentiment analysis, sensitivity analysis, and dashboard development.</i> <i>Transparent ranking logic where users can see the indicators, weights, score components, and data completeness.</i>
Out of Scope	<i>Ranking every sustainability Master’s program in the world, because the global population is too large and public data is inconsistent.</i> <i>Verifying private university claims such as exact employment outcomes unless reliable public evidence is available.</i> <i>Treating Reddit comments as verified facts. Reddit will be used only as a peer-perception signal.</i> <i>Collecting private student data, identifying Reddit users, or using non-public communities.</i>

5 Data Sources

Source	Type / Format	Access & Availability	Notes
<i>Program official websites</i>	<i>HTML</i>	<i>Public; to be scraped or manually collected</i>	<i>Public Information from the official website including key aspects and their own feature toward the program</i>

Source	Type / Format	Access & Availability	Notes
QS World University Rankings: Sustainability	HTML https://www.topuniversities.com/sustainability-rankings	Public	Sustainability rank, available sustainability scores or indicators
Times Higher Education Impact Rankings	HTML https://www.timeshighereducation.com/impactrankings	Public; to be scraped or manually collected	Detailed data may be limited
AASHE STARS	Tables(csv) https://reports.aashe.org/institutions/participants-and-reports/	Public; to be scraped or manually collected	Transparent self-reported sustainability performance data for higher education institutions.
Reddit	JSON	Public; to be scraped	Analyzing comments by students who have graduated from the program
UN PRME Signatory Members	Web directory https://www.unprme.org/signatory-members/	Public	Identifies business and management schools committed to responsible management education.
OpenAlex	API https://openalex.org	Public API	Estimate sustainability-related research output, citations, SDG-related publications, and research intensity by institution.

Source	Key Fields Used	Current / Expected Row Count	Quality Assessment
Official program websites	Program name, university, country, degree type, duration, tuition, curriculum, language, delivery mode, internship, capstone, thesis option	Target: 50–100 program rows	Most relevant source because it is program-level, but formatting is inconsistent and some information may require manual validation.
QS World University	Institution name, country, sustainability rank, available sustainability indicators	Expected matched rows: 30–70	Useful institutional sustainability signal, but not program-level. It will be

Source	Key Fields Used	Current / Expected Row Count	Quality Assessment
Times Higher Education Impact Rankings	Institution name, country, overall impact rank, SDG-related indicators where available	Expected matched rows: 30–70	used as an input feature rather than the final ranking. Strong SDG-related institutional signal, but detailed indicators may be limited and may not directly describe specific Master's programs.
AASHE STARS	Institution name, country, STARS rating, report year, sustainability performance categories	Expected matched rows: 10–40	Transparent and detailed, but coverage is stronger for North American institutions and data is self-reported.
UN PRME Signatory Members	Institution name, country, signatory status, responsible management education commitment	Expected matched rows: 20–60	Useful for business and management schools, but signatory status is not a direct program-quality measure.
OpenAlex	Institution name, sustainability-related publications, citations, SDG-related research output	Expected matched rows: 30–70 institution rows	Open research data source, but institution matching requires cleaning and may not perfectly represent the specific Master's program.
Reddit	Program mentions, university mentions, post/comment text, subreddit, date, score, sentiment, discussion themes	Expected raw text rows: 100–1,000 before cleaning	Valuable for peer perception, but noisy, self-selected, uneven across programs, and not representative. It will be aggregated and confidence-flagged.

Data Readiness and Backup Plan

Before finalizing the full dataset, the team will create a pilot sample of 10–15 programs and test whether each source provides usable fields, matched institutions, row counts, and sample records. The team will record sample columns such as program name, tuition, duration, country, curriculum focus, sustainability rank, research output, Reddit discussion count, and sentiment label. If a source has insufficient coverage, it will be downgraded to a supplementary feature or replaced with manual collection from official university pages. If Reddit discussion is too sparse for program-level analysis, Reddit will be aggregated at the university level and reported with a low peer-data confidence score rather than used as a direct penalty.

6 Methodology

The project will use a multi-stage data pipeline moving from program discovery to extraction, cleaning, text analysis, scoring, ranking, validation, and dashboard development. The main tools will be Python, pandas, BeautifulSoup, Reddit API access where feasible, natural language processing libraries, Excel, and a dashboard tool such as Tableau, or Power BI. R may be used for statistical checks if needed.

First, the team will identify sustainability-focused Master's programs using keywords such as "sustainability," "sustainable development," "climate," "ESG," "environmental management," "sustainable finance," "responsible management," "social impact," and "sustainable business." The initial sample will prioritize business and management programs with public information, geographic diversity, and enough available data to support meaningful comparison.

Second, structured program data will be collected from official websites, brochures, tuition pages, and curriculum pages. The team will standardize university names, countries, tuition currencies, degree duration, delivery mode, and program features. Program attributes will be transformed into comparable indicators using binary flags, categorical scoring, and normalized numeric scores. For example, internship or capstone availability may be coded as binary variables, while tuition and research-output variables may be normalized before scoring.

Third, Reddit posts and comments will be collected as a peer-evaluation source. Search queries will combine program names, university names, and terms such as "worth it," "tuition," "job prospects," "admissions," "student experience," and "sustainability master." Duplicate, irrelevant, deleted, or spam-like content will be removed.

Username will not be reported. Sentiment analysis will classify discussion as positive, negative, or neutral, while topic analysis will identify themes such as cost, career outcomes, workload, location, admissions difficulty, and perceived program value. Because Reddit is biased and incomplete, it will be used as a modest peer-perception signal rather than verified evidence. Programs with limited Reddit discussion will receive a lower peer-data confidence score instead of being penalized.

Fourth, the team will build a weighted ranking model. The baseline weights will be: curriculum relevance 25%, affordability 20%, experiential learning 15%, institutional sustainability credibility 15%, research strength 10%, career signals 10%, and Reddit peer sentiment 5%. Alternative weighting scenarios will be tested for cost-sensitive, research-oriented, career-oriented, impact-oriented, and peer-review-sensitive students.

Fifth, validation will include sensitivity analysis, outlier checks, missing-data review, and manual review of a sample of Reddit sentiment classifications. The result will be considered successful if the final dataset includes at least 50 programs, reaches at least 75% average completeness across required fields, supports at least five dashboard filters, and tests at least three alternative weighting scenarios. Skill gaps are mainly related to advanced web scraping, especially when program pages use inconsistent layouts, dynamic content, or access restrictions. The team will address this through documentation, office hours, and backup methods such as manual collection from official program pages and brochures.

7 Deliverables

#	Deliverable	Description	Format
1	Final report	Explains the problem, data sources, methodology, ranking framework, results, limitations	PDF
2	Clean program database	Structured dataset of selected sustainability Master's programs with standardized fields and data-source notes	Excel
3	Searchable ranking prototype	Interactive tool that allows users to search, filter, compare programs, and adjust weights, including whether to include peer sentiment	Dashboard

8 Timeline / Gantt

Phase / Task	W1	W2	W3	W4	W5	W6	W7	W8
Define & scope								
Data collection			Progress 1					
Analysis / build				Progress 2				
Validation						Progress 3		
Write-up & deliverables							Progress 4	Final Pre

Week	Dates	Milestones
Week 1	June 1– June 7	Confirm project scope, finalize title, define target program categories, assign roles, and create the first draft of the data dictionary.
Week 2	June 8– June 14	Build the initial program list, test official website extraction, and identify matching fields for QS, THE, AASHE, UN PRME, OpenAlex, and Reddit.
Week 3	June 15– June 21	Collect program website data and begin matching programs with institutional ranking and sustainability sources. Submit Progress Report 1 .

Week	Dates	Milestones
Week 4	June 22– June 28	Collect OpenAlex research-output data, design Reddit search queries, and clean the first version of the program database.
Week 5	June 29– July 5	Complete the initial Reddit data collection, clean text data, and begin sentiment and topic analysis. Submit Progress Report 2 .
Week 6	July 6–July 12	Normalize variables, calculate preliminary scores, create missing-data flags, and draft the ranking methodology.
Week 7	July 13– July 19	Build the dashboard prototype, add search and filter functions, add ranking-weight logic, and conduct the first sensitivity analysis. Submit Progress Report 3 .
Week 8	July 20– July 26	Validate rankings, manually review a sample of Reddit sentiment results, finalize the dashboard, and draft the final report. Submit Progress Report 4 .
Final Week	July 27– July 30	Finalize the report, clean the database, prepare presentation slides, rehearse, and complete the Final Presentation .

The project includes buffer time in Weeks 3, 5, 7, and 8. Week 3 allows extra time for website-scraping problems and manual collection if official program pages are difficult to parse. Week 5 allows flexibility for Reddit API limitations or low discussion volume. Week 7 includes buffer for dashboard debugging and ranking-weight adjustments. Week 8 is reserved for validation, manual checks, final edits, and fixing unexpected data-quality problems before the final presentation.

9 Risks

Risk	Likelihood	Impact	Mitigation
<i>Reddit discussion volume is low for some programs</i>	<i>H</i>	<i>M</i>	<i>Use Reddit only when enough discussion exists; include a peer-data confidence score instead of penalizing missing discussion</i>
<i>Ethical or privacy concerns arise from user-generated content</i>	<i>M</i>	<i>H</i>	<i>Use only public content, avoid usernames, aggregate results, and do not quote sensitive individual comments in the final ranking</i>
<i>Program-level data is incomplete or inconsistent across universities</i>	<i>H</i>	<i>H</i>	<i>Use data-completeness flags, manual validation for key fields, and avoid over-penalizing missing data</i>
<i>Weighting choices may appear subjective</i>	<i>M</i>	<i>M</i>	<i>Provide a clear rationale, multiple weighting scenarios, and sensitivity analysis</i>
<i>Employment outcome data may be unavailable or biased</i>	<i>H</i>	<i>M</i>	<i>Treat career outcomes as optional or low-weight indicators; use public signals cautiously</i>
<i>Web scraping is blocked or pages are difficult to parse</i>	<i>M</i>	<i>L</i>	<i>Use a hybrid approach: scraping for structured fields and manual collection for complex pages</i>

10 Roles

Team Member	Primary Responsibilities	Backup For
<i>Yuyang Chen</i>	<i>Project management, final report structure, dashboard or searchable database prototype, user interface, visualization</i>	<i>Data collection</i>

Team Member	Primary Responsibilities	Backup For
Zhaihan Gong	Data cleaning, normalization, data dictionary, missing-data flags	Dashboard
Alan Yang	Ranking model, weighting framework, sensitivity analysis, validation	Final report and presentation

11 Success Metrics.

Metric	Definition	Target	How Measured
Program coverage	Number of sustainability Master's programs included in the final database	At least 50 programs	Count of cleaned program records
Dashboard usability	Ability to filter by country, region, cost, program type, duration, sustainability focus, and peer sentiment	At least 4 useful filters	Dashboard
User usefulness	Whether the ranking helps students compare programs more easily than existing rankings	Feedback from sample users or class review	Short feedback survey
Ranking robustness	Reasonable ranking rule with comment and introduction to each program	All program ranking makes sense	Sensitivity analysis

AI disclosure --- the words and paragraphs are optimized by AI. Timeline is suggested by AI.